

2025

WFIGS EXPORT TOOL - USER MANUAL & TECHNICAL GUIDE



WASHINGTON STATE DEPARTMENT OF
NATURAL RESOURCES
WILDLAND FIRE MANAGEMENT
DIVISION - AVIATION OPERATIONS
MULTI MISSION AIRCRAFT (MMA)
PROGRAM

Serfass, Joshua (DNR)

8/31/2025

Version 2.0

TABLE OF CONTENTS

Introduction.....	3
Section I - Tool Overview.....	4
System Requirements.....	4
Installation.....	5
What's New in Version 2.0.....	6
Section II - User Interface Guide.....	7
Parameter Overview.....	7
Layer Selection Parameters.....	8
Geographic and Filter Parameters.....	10
Output Configuration Parameters.....	12
Symbology Parameters.....	14
Section III - Workflow Instructions.....	16
Basic Workflow.....	16
Advanced Configuration.....	19
Processing Flow Diagram.....	21
Section IV - Output Files and Data Management.....	23
File Naming Conventions.....	23
Output Types and Formats.....	24
Quality Control.....	26
Section V - Troubleshooting and Optimization.....	28
Common Issues.....	28
Performance Optimization.....	30
Best Practices.....	31
Section VI - Technical Reference.....	33
Code Architecture.....	33
Service Endpoints and Data Sources.....	34
Configuration Details.....	36
Maintenance and Updates.....	37
Glossary.....	38
Appendix A - Service URL Reference.....	39
Appendix B - Change Log.....	40

INTRODUCTION

What is the WFGIS Export Tool?

The WFGIS Export Tool is a specialized ArcGIS Python toolbox developed specifically for Washington State Department of Natural Resources (DNR) Aviation Operations. This automated solution streamlines the export of critical wildfire GIS data from multiple sources including the Wildland Fire Geospatial Interagency (WFIGS) system, VIIRS satellite thermal detection, and Interagency Aviation Agreement (IAA) mission systems, converting these diverse data layers into KMZ files optimized for Google Earth and field operations platforms.

Version 2.0 introduces significant enhancements including selective layer export capabilities, thermal hotspot integration, and IAA mission area support, providing unprecedented flexibility in data management and operational intelligence gathering.

Tool Capabilities and Mission Support

The WFGIS Export Tool enhances operational effectiveness by:

- **Selective Layer Processing:** Choose specific data layers for export based on operational needs
- **Lightning Intelligence:** Automated download and processing of 24-hour and 72-hour lightning strike data
- **Thermal Detection:** Integration of VIIRS satellite thermal hotspot data for early fire detection
- **Active Incident Tracking:** Real-time retrieval of fire incident locations with current status information
- **Perimeter Mapping:** Obtaining official fire perimeter polygons from authoritative sources
- **IAA Mission Support:** Dedicated processing of Interagency Aviation Agreement areas of interest with incident-specific filtering
- **Intelligent Symbolology:** Automatic application of DNR-standard symbology for immediate field interpretation
- **Flexible Export Options:** KMZ creation with automatic shapefile fallback for compatibility
- **Comprehensive Logging:** Detailed processing logs for quality assurance and troubleshooting

Target Users and Applications

Primary users include:

- DNR Aviation Personnel conducting reconnaissance and tactical missions
- Fire Operations Coordinators requiring current multi-source intelligence
- GIS Analysts supporting incident management teams

- Emergency Response Teams needing field-ready data packages
- Incident Management Teams (IMTs) planning strategic operations
- Interagency Aviation Coordinators managing resource deployment

Operational scenarios supported:

- Initial attack intelligence gathering with selective data layers
 - Extended campaign support with comprehensive data packages
 - Thermal detection monitoring for emerging fire activity
 - IAA mission planning and area of interest tracking
 - Cross-border operations with multi-state data integration
 - Real-time situational awareness for aerial operations
-

SECTION I - TOOL OVERVIEW

System Requirements

Software Requirements

- ArcGIS Pro (Version 2.8 or higher recommended, 3.0+ preferred)
- ArcGIS Desktop (Version 10.7 or higher, if using ArcMap - limited support)
- Python 3.x (included with ArcGIS Pro installation)
- Windows 10 or higher operating system (Windows 11 supported)

Network Requirements

- Reliable internet connection for accessing multiple web services
- Firewall access to the following domains: - services3.arcgis.com (WFIGS services) - services9.arcgis.com (VIIRS thermal data)
- Minimum 10 Mbps download speed (25 Mbps recommended for full dataset processing)
- Network stability for sustained connections during large data transfers

Hardware Requirements

- **RAM:** Minimum 8GB (16GB recommended, 32GB optimal for processing all layers simultaneously)
- **Storage:** Minimum 5GB available space for temporary files and outputs
- **Processor:** Multi-core processor recommended (4+ cores for optimal performance)
- **Display:** Minimum 1920x1080 resolution for proper interface viewing

Installation

Step 1: Obtain the Tool

1. Download the WFIGS_Export_Tool.pyt file from your DNR system administrator

2. Verify file integrity (current version should be approximately 35-40KB)
3. Create a dedicated tools directory (recommended: C:_GIS_Tools_v2)
4. Place the tool file in the designated directory
5. Document the installation date and version for maintenance tracking

Step 2: Add to ArcGIS Pro

6. Launch ArcGIS Pro with a map project
7. Open the Catalog pane (View > Catalog Pane)
8. Navigate to Toolboxes in the catalog tree
9. Right-click Toolboxes and select “Add Toolbox”
10. Browse to the WFIGS_Export_Tool.pyt file location
11. Select the file and click “OK”
12. Verify the toolbox appears as “WFIGS Export Tools”
13. Expand to confirm “WFIGS Export Tool” is available

Step 3: Verify Installation

14. Double-click “WFIGS Export Tool” to open the interface
15. Confirm all 17 parameters are visible and properly labeled
16. Verify layer selection checkboxes appear at the top
17. Check that default paths are populated in symbology parameters
18. Test the IAA AOI Filter dropdown population (if Export IAA is checked)
19. Close the tool dialog without running

Step 4: Configure Default Paths

20. Verify access to default symbology directory:
J:_opsProgram\7-Chief PilotWing (FW)\GIS\2025\2025_kml_file_conversionsFiles
21. Confirm the following symbology files exist:
 - 24h_Lightning.lyrx
 - 72h_Lightning.lyrx
 - Thermal_Hotspots.lyrx
 - WFIGS_Incidents.lyrx
 - WFIGS_Perimeters.lyrx
 - IAA_2025_Mission_Request.lyrx
22. If default paths are inaccessible, prepare custom symbology files
23. Test network connectivity to all service endpoints (see Appendix A)

What's New in Version 2.0

Major Enhancements

- **Selective Layer Export:** New checkbox interface allows users to select specific layers for processing, reducing processing time and file management overhead
- **Thermal Hotspot Integration:** Added support for VIIRS satellite thermal detection data, providing early fire detection capability
- **IAA Mission Support:** New IAA Area of Interest layer with dynamic incident filtering for targeted mission planning
- **Enhanced User Interface:** Reorganized parameter layout with logical grouping and intelligent parameter enabling/disabling
- **Improved Error Handling:** Multi-tier fallback mechanisms ensure data export even when primary methods fail

New Data Layers

- Thermal Hotspots from VIIRS satellite constellation
- IAA Area of Interest with mission-specific filtering
- Dynamic incident name filtering for IAA operations

Performance Improvements

- Optimized processing for individual layer selection
- Reduced memory usage through selective data loading
- Enhanced cleanup procedures to prevent resource conflicts
- Improved KML export parameters specific to each layer type

Quality of Life Updates

- All layers default to selected for backward compatibility
- Automatic validation ensures at least one layer is selected
- Enhanced logging shows which layers were processed
- Improved file naming includes incident names for IAA exports

SECTION II - USER INTERFACE GUIDE

Parameter Overview

The WFGIS Export Tool interface contains 17 parameters organized into logical groups for efficient workflow management:

Layer Selection Controls (Parameters 1-6)

- Export 24h Lightning: Toggle for 24-hour lightning data processing
- Export 72h Lightning: Toggle for 72-hour lightning data processing
- Export Incidents: Toggle for active fire incident points
- Export Perimeters: Toggle for fire boundary polygons
- Export Thermal Hotspots: Toggle for VIIRS satellite thermal detection
- Export IAA Area of Interest: Toggle for interagency aviation mission areas

Geographic Configuration (Parameters 7-8)

- State Selection: Defines the processing area for incidents and perimeters
- IAA AOI Filter: Dynamic dropdown for incident-specific filtering

Spatial Filtering (Parameter 9)

- Mask/Clip Feature: Optional boundary for lightning and thermal data

Output Management (Parameters 10-11)

- Output Folder: Destination directory for all exported files
- Max Lightning Features: Feature count limit for performance management

Symbology Configuration (Parameters 12-17)

- Individual symbology paths for each data layer type
- Support for both .lyrx (ArcGIS Pro) and .lyr (legacy) formats

Layer Selection Parameters

Export 24h Lightning

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Enables mask parameter when selected

Technical Implementation: When selected, the tool connects to the WFIGS 24-hour lightning service and retrieves all strikes from the past 24 hours. The data undergoes spatial filtering if a mask is provided and feature limiting based on the max features parameter.

Operational Considerations:

- Essential for initial attack intelligence
- Typically contains 1,000-50,000 features depending on storm activity
- Processing time increases significantly during active weather patterns
- Consider unchecking during quiet weather periods to reduce processing time

Export 72h Lightning

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Enables mask parameter when selected

Technical Implementation: Retrieves lightning strikes from the past 72 hours, providing extended temporal context for fire cause determination and pattern analysis. Uses the same spatial filtering and limiting logic as 24-hour data.

Use Cases:

- Fire investigation and cause determination
- Extended weather pattern analysis
- Historical context for new fire starts
- Uncheck when only current intelligence is needed

Export Incidents

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Requires state parameter selection

Technical Implementation: Queries the WFIGS incident database with complex SQL filtering to retrieve active fires meeting specific criteria:

- GACC = 'NWCC' (Northwest Coordination Center)
- Fire size ≥ 1 acre
- Not controlled, contained, or out
- Discovered within last 30 days
- Includes both wildfires (WF) and prescribed burns (RX)

Data Attributes Preserved:

- IncidentName: Used for labeling in Google Earth
- POOState: Point of origin state
- IncidentSize: Current reported size
- FireDiscoveryDateTime: Initial report time
- All original WFIGS attributes for analysis

Export Perimeters

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Requires state parameter selection

Technical Implementation: Retrieves current fire perimeter polygons for fires greater than 50 acres. The tool applies special KML export parameters to preserve transparency and reduce edge blurring in Google Earth visualization.

Visualization Optimizations:

- Reduced composite rendering to prevent blur
- Higher resolution image export (1024px)
- Optimized scale thresholds for clarity
- Automatic fallback to minimal parameters if initial export fails

Export Thermal Hotspots

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Enables mask parameter when selected

Data Source: VIIRS satellite constellation

Technical Implementation: Connects to the services9.arcgis.com VIIRS thermal detection service to retrieve current satellite-detected heat anomalies. This data provides early detection capability for emerging fires and validation of ground reports.

Key Features:

- Near real-time satellite detection (typically 3-4 hour latency)
- Global coverage with 375m resolution
- Confidence ratings for each detection
- Temporal filtering to focus on recent detections
- Spatial filtering support through mask parameter

Operational Value:

- Early fire detection in remote areas
- Validation of smoke reports
- Monitoring of fire growth patterns
- Detection of holdover fires from lightning

Export IAA Area of Interest

Type: Boolean checkbox

Default: Checked (True)

Dependencies: Enables IAA AOI Filter parameter

Technical Implementation: Queries the 2025 IAA mission request system for areas assigned to specific aircraft (N565E and N216KQ) with “Assigned” mission status. Supports dynamic filtering by incident name through the dropdown interface.

Mission Integration Features:

- Real-time mission area updates
- Aircraft-specific filtering

- Incident name-based selection
- Preserves all mission attributes including: - Mission_Status - Incident_Name - Assigned_To (aircraft identifier) - Mission request details

File Naming Convention:

- Includes incident name in output filename
- Format: YYYYMMDD_IAA_AOI_IncidentName_State.kmz
- “AllIncidents” tag when no specific filter applied

Geographic and Filter Parameters

State (for Incidents & Perimeters)

Type: Required dropdown selection

Options: WA, OR, Both

Default: WA (Washington)

Affects: Incidents and Perimeters layers only

Technical Implementation: Constructs optimized SQL WHERE clauses for server-side filtering:

- WA: POOState = ‘US-WA’
- OR: POOState = ‘US-OR’
- Both: POOState IN (‘US-WA’, ‘US-OR’)

Performance Considerations:

- Single state queries typically return in 5-10 seconds
- “Both” selection may double processing time
- Server-side filtering reduces network traffic by 60-80%
- Consider operational area when selecting scope

Best Practices:

- Use “WA” for routine DNR operations
- Select “OR” for mutual aid deployments
- Choose “Both” for: - Columbia River Gorge fires - Regional situation awareness - Multi-state campaign operations - Comprehensive intelligence products

IAA AOI Filter

Type: Dynamic dropdown

Default: “All Incidents”

Availability: Only enabled when Export IAA is checked

Population: Dynamically queries available incidents on parameter update

Technical Implementation: The tool performs real-time queries to the IAA service to populate the dropdown with current incident names. When a specific incident is selected, it adds an additional SQL filter to retrieve only that incident's assigned areas.

Dynamic Behavior:

- Refreshes each time the tool dialog is opened
- Updates when Export IAA checkbox is toggled
- Displays “No incidents found” if query returns empty
- Shows “Error loading incidents” if connection fails
- Maintains selection between parameter updates

Filter Application:

- “All Incidents”: Shows all assigned missions for configured aircraft
- Specific incident: Filters to selected incident's areas only
- Applied at server level for optimal performance
- Reflected in output filename for clear identification

Use Cases:

- Individual incident briefing products
- Mission-specific planning maps
- Focused area reconnaissance
- Reduced clutter in complex incidents

Mask/Clip Feature (for Lightning & Thermal Hotspots only)

Type: Optional feature layer selection

Source: Active layers in current ArcGIS Pro project

Supported Geometry: Polygon features only

Affects: Lightning (24h and 72h) and Thermal Hotspots only

Technical Implementation: Performs client-side spatial intersection using ArcGIS SelectLayerByLocation with “INTERSECT” overlap type. The tool maintains source data projection throughout processing to ensure spatial accuracy.

Spatial Processing Details:

- Intersection calculated after initial data retrieval
- Preserves all attributes of filtered features
- No buffer applied (exact intersection only)
- Handles multi-part polygons correctly
- Supports complex boundaries with holes

Performance Impact:

- Small masks (<100 vertices): Minimal impact
- Complex masks (>1000 vertices): 20-30% slower
- Very complex masks (>10000 vertices): Consider simplification
- Large area masks: Limited performance benefit

Recommended Mask Sources:

- Dispatch zones for operational boundaries
- Fire Management Units for planning
- County boundaries for jurisdictional filtering
- Custom AOI polygons for specific operations
- Watershed boundaries for resource protection
- WUI boundaries for community protection focus

Important Notes:

- Does NOT affect Incidents, Perimeters, or IAA layers
- Mask must be loaded in current map before tool execution
- Coordinate system mismatches handled automatically
- Null or invalid geometries in mask will cause errors

Output Configuration Parameters

Output Folder

Type: Required folder browser

Default: J:_opsProgram\7-Chief PilotWing

(FW)GIS\2025\2025_kml_file_conversions_Files_Daily_Exports

Permissions: Write access required

Recommended: Network location for shared access

Technical Implementation: All outputs are written to this directory with standardized naming conventions. The tool creates a comprehensive log file alongside data exports, providing complete processing documentation.

Directory Organization Recommendations:

```
WFIGS_Daily_Exports/  
├── 2025/  
│   ├── January/  
│   ├── February/  
│   └── ...  
├── Archives/  
├── IAA_Specific/  
└── Emergency_Exports/
```

Storage Requirements:

- Minimum 500MB free space for standard export
- 1-2GB recommended during high activity periods
- Consider automated cleanup of files >30 days old
- Network drives should have stable connections
- Local SSD provides 3-5x faster processing

File Management Features:

- Automatic timestamp integration prevents overwrites
- Log files provide processing audit trail
- Shapefile fallback creates subdirectories
- No automatic deletion of existing files

Max Lightning Features

Type: Optional integer input

Default: 15,000 features

Range: 0 (no limit) to 100,000

Special: 0 value removes all limitations

Affects: Both 24h and 72h lightning layers

Technical Implementation: Implements intelligent feature limiting using database cursors to process large datasets without loading entire feature sets into memory. Features are selected based on original query order (typically temporal).

Memory Management Strategy:

- Streaming cursor implementation for large datasets
- Features processed in 1000-record batches
- Memory usage remains constant regardless of total count
- No impact on spatial accuracy or attribute completeness

Recommended Settings by Use Case:

- **Initial Attack Intelligence:** 5,000-8,000 features - Focuses on most recent strikes - Reduces file size for mobile devices - Faster processing and transfer
- **Standard Operations:** 15,000 features (default) - Balanced coverage and performance - Suitable for most operational needs - Manageable file sizes (2-5MB KMZ)
- **Comprehensive Analysis:** 25,000-50,000 features - Extended temporal coverage - Pattern analysis capability - Larger files (5-15MB KMZ)
- **No Limit (0):** Use with caution - Complete dataset retrieval - May result in very large files - Processing times can exceed 30 minutes - Risk of memory errors on limited systems

Performance Benchmarks:

- 5,000 features: ~2 minutes processing
- 15,000 features: ~5-8 minutes processing
- 50,000 features: ~15-20 minutes processing
- Unlimited: Highly variable (10-60+ minutes)

Symbology Parameters

All symbology parameters share common characteristics:

- **Type:** Optional file browser
- **Supported formats:** .lyrx (ArcGIS Pro) and .lyr (ArcGIS Desktop)
- **Fallback:** Tool continues with default styling if files are missing
- **Validation:** File existence checked before application

24h Lightning Symbology

Default: 24h_Lightning.lyrx

Standard Configuration:

- Yellow to red color ramp based on age
- Smaller symbols for older strikes
- Semi-transparent for overlay visibility
- Labels showing strike time (optional)

72h Lightning Symbology

Default: 72h_Lightning.lyrx

Standard Configuration:

- Blue to yellow color ramp for extended timeline
- Size gradation by age (older = smaller)
- Lower opacity for temporal context
- Clustering at small scales

Thermal Hotspots Symbology

Default: Thermal_Hotspots.lyrx

Standard Configuration:

- Red to orange gradient by confidence level
- Larger symbols for high-confidence detections
- Timestamp-based transparency
- Labels showing detection time and confidence

Key Elements:

- Confidence-based sizing (High/Medium/Low)
- Recency-based transparency
- Clear differentiation from fire incidents
- Optimized for overlay with other fire data

Incidents Symbolology

Default: WFIGS_Incidents.lyrx

Standard Configuration:

- Red flame symbols for active fires
- Size based on incident acreage
- Labels showing incident names
- Different symbols for WF vs RX types

Label Configuration for Google Earth:

- IncidentName field for primary labels
- Font size scales with zoom level
- High contrast colors for visibility
- Automatic decluttering at small scales

Perimeters Symbolology

Default: WFIGS_Perimeters.lyrx

Standard Configuration:

- Semi-transparent red fill
- Bold red outline
- Transparency preserved in KML export
- Labels showing fire names and acres

Special Export Settings:

- Reduced blur through non-composite rendering
- Higher resolution for edge clarity
- Optimized transparency (40-60% typical)
- Z-order places below point features

IAA AOI Symbolology

Default: IAA_2025_Mission_Request.lyrx

Standard Configuration:

- Mission_Status field drives symbolization
- Assigned areas in bright green

- Completed missions in gray
- Pending requests in yellow

Attribute-Driven Styling:

- Field: Mission_Status
- Values: Assigned, Completed, Pending
- Unique symbols for each aircraft
- Labels show incident name and aircraft

Custom Symbology Development:

1. Create styled layer in ArcGIS Pro
 2. Configure all visual properties
 3. Save as layer file (.lyrx)
 4. Place in accessible directory
 5. Test with sample data
 6. Update tool parameter paths
-

SECTION III - WORKFLOW INSTRUCTIONS

Basic Workflow

Pre-Execution Checklist

1. Verify ArcGIS Pro is running with appropriate permissions
2. Confirm internet connectivity to all service endpoints
3. Check available disk space (minimum 500MB recommended)
4. Load any required mask features into the current map
5. Close unnecessary applications to maximize resources
6. Ensure network drives are mapped and accessible

Step 1: Launch the Tool

1. Open ArcGIS Pro with an active map project
2. Navigate to Catalog > Toolboxes > WFIGS Export Tools
3. Double-click “WFIGS Export Tool” to open the interface
4. Wait for parameter initialization (10-15 seconds typical)
5. Observe IAA dropdown population if applicable

Technical Note: Initial parameter validation includes network connectivity tests, service availability checks, and dropdown population for dynamic parameters.

Step 2: Select Layers for Export

1. Review the six layer checkboxes at the top of the interface
2. Check/uncheck layers based on operational needs:
 - ☒ Export 24h Lightning - Recent strike activity
 - ☒ Export 72h Lightning - Extended strike history
 - ☒ Export Incidents - Active fire points
 - ☒ Export Perimeters - Fire boundaries
 - ☒ Export Thermal Hotspots - Satellite heat detection
 - ☒ Export IAA Area of Interest - Mission assignments
3. Ensure at least one layer is selected (tool validates this)
4. Note which parameters become enabled/disabled based on selection

Layer Selection Strategies:

- **Rapid Intelligence:** Lightning 24h + Incidents only (2-3 min)
- **Standard Package:** All except IAA (5-10 min)
- **Comprehensive:** All layers selected (10-15 min)
- **IAA Mission Focus:** IAA + Incidents + Perimeters (3-5 min)
- **Fire Weather:** Both lightning + Thermal hotspots (5-8 min)

Step 3: Configure Geographic Parameters

1. Select target state from dropdown (WA, OR, or Both)
 - Affects only Incidents and Perimeters
 - Lightning and Thermal data are not state-filtered
2. If Export IAA is checked:
 - Wait for AOI Filter dropdown to populate
 - Select “All Incidents” or specific incident name
 - Dropdown refreshes each time dialog is opened
3. If using mask feature:
 - Ensure polygon layer is loaded in current map
 - Select from dropdown (only polygon layers appear)
 - Verify mask covers area of interest

Geographic Considerations:

- State selection drives server-side filtering for efficiency
- Mask filtering occurs client-side after data retrieval
- IAA filtering is independent of state selection
- Consider operational boundaries when configuring

Step 4: Set Output Configuration

1. Browse to select output directory
 - Default location recommended for standard operations
 - Custom location for special exports
2. Verify write permissions to selected folder
3. Set Max Lightning Features:
 - Leave at 15,000 for standard operations
 - Reduce to 5,000 for mobile/bandwidth limited
 - Increase to 25,000+ for analysis products
 - Set to 0 only when complete dataset required

Output Planning:

- Network locations enable team access
- Local drives provide faster processing
- Consider creating date-based subdirectories
- Plan for 10-50MB per complete export set

Step 5: Verify Symbology (Optional)

1. Review default symbology paths
 - Tool pre-populates standard DNR symbology
 - Files are validated during execution
2. Browse to custom symbology if needed:
 - Must be .lyrx or .lyr format
 - Should match data schema
 - Test custom symbology before operational use
3. Clear parameter to use default styling
 - Tool continues without error if files missing
 - Basic default symbols applied automatically

Symbology Best Practices:

- Use standard DNR symbology for consistency
- Create incident-specific symbology as needed
- Maintain symbology library for common scenarios
- Document custom symbology configurations

Step 6: Execute Processing

1. Review all parameters for accuracy
2. Click “Run” to begin processing
3. Monitor Messages window for progress:

```
Starting WFGIS Export Tool...
Selected layers for export: 24h Lightning, Incidents, Perimeters
[Lightning24h] Processing...
[Lightning24h] Spatially filtered count: 8,432
[Lightning24h] Successfully exported KMZ: 20250829_Lightning24h_WA.kmz
[Incidents] Processing...
[Incidents] State filtered count: 47
[Incidents] Successfully exported KMZ: 20250829_Incidents_WA.kmz
```

4. Do not close ArcGIS Pro during processing
5. Processing completes when log summary appears

Expected Processing Times by Configuration:

- Single layer: 1-3 minutes
- Lightning + Incidents: 3-5 minutes
- All layers except IAA: 8-12 minutes
- All layers with mask: 12-18 minutes
- All layers, Both states: 15-25 minutes

Step 7: Verify Results

1. Navigate to output folder
2. Verify expected files are present:
 - KMZ files for each selected layer
 - Log file with timestamp
 - Shapefile folders if KMZ export failed
3. Open log file to review processing details:
 - Feature counts for each layer
 - Any warnings or errors
 - Total processing time
4. Test KMZ files in Google Earth:
 - Verify symbols display correctly
 - Check labels are visible (Incidents)
 - Confirm perimeter transparency
 - Test feature popups for attributes

Quality Verification Checklist:

- ☐ All selected layers have output files
- ☐ File sizes are within expected ranges

- ☐ Log file shows no critical errors
- ☐ KMZ files open in Google Earth
- ☐ Symbology displays as expected
- ☐ Feature counts match operational expectations
- ☐ Spatial extent covers area of interest

Advanced Configuration

Rapid Initial Attack Configuration

Optimized for time-critical response with minimal processing:

Layer Selection: - ☒ Export 24h Lightning - ☒ Export Incidents - ☐ Export 72h Lightning - ☐ Export Perimeters - ☒ Export Thermal Hotspots - ☐ Export IAA Area of Interest

Parameters: - State: Specific state of interest - Mask: Dispatch zone or response area - Max Lightning: 5,000 features - Output: Local SSD for speed

Processing Time: 2-4 minutes

Output Size: 5-10MB total

Use Cases: • New start reconnaissance

- Initial size-up intelligence
- Resource positioning decisions
- Cause determination support

Comprehensive Campaign Support

Full dataset for extended operations and planning:

Layer Selection: - ☒ All layers selected

Parameters: - State: Both (for regional awareness) - IAA Filter: All Incidents - Mask: None (full coverage) - Max Lightning: 25,000 features - Custom symbology for specific incident

Processing Time: 20-30 minutes

Output Size: 30-50MB total

Use Cases: • Daily intelligence products

- Briefing package generation
- Strategic planning support
- Interagency coordination

IAA Mission-Specific Export

Focused export for assigned aviation missions:

Layer Selection: - ☐ Export 24h Lightning - ☐ Export 72h Lightning - ☒ Export Incidents - ☒ Export Perimeters - ☐ Export Thermal Hotspots - ☒ Export IAA Area of Interest

Parameters: - State: As appropriate for mission - IAA Filter: Specific incident name - Output: Mission-specific folder

Processing Time: 3-5 minutes

Output Size: 10-15MB

File Naming: 20250829_IAA_AOI_RidgeFire_WA.kmz

20250829_Incidents_WA.kmz

20250829_Perimeters_WA.kmz

Multi-User Environment Configuration

Preventing Conflicts: 1. Establish processing schedules 2. Use user-specific output folders 3. Implement file naming conventions with user ID 4. Coordinate layer selection to avoid redundancy

Shared Resource Setup:

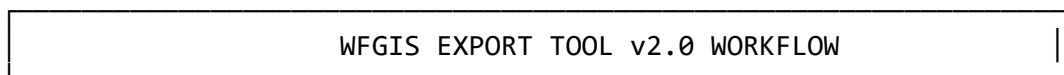
Network Structure:

```
\\server\WFGIS_Exports\  
├── User_Exports\  
│   ├── User1\  
│   ├── User2\  
│   └── User3\  
├── Daily_Products\  
│   └── YYYYMMDD\  
├── Symbology\  
│   └── Standard\  
└── Logs\
```

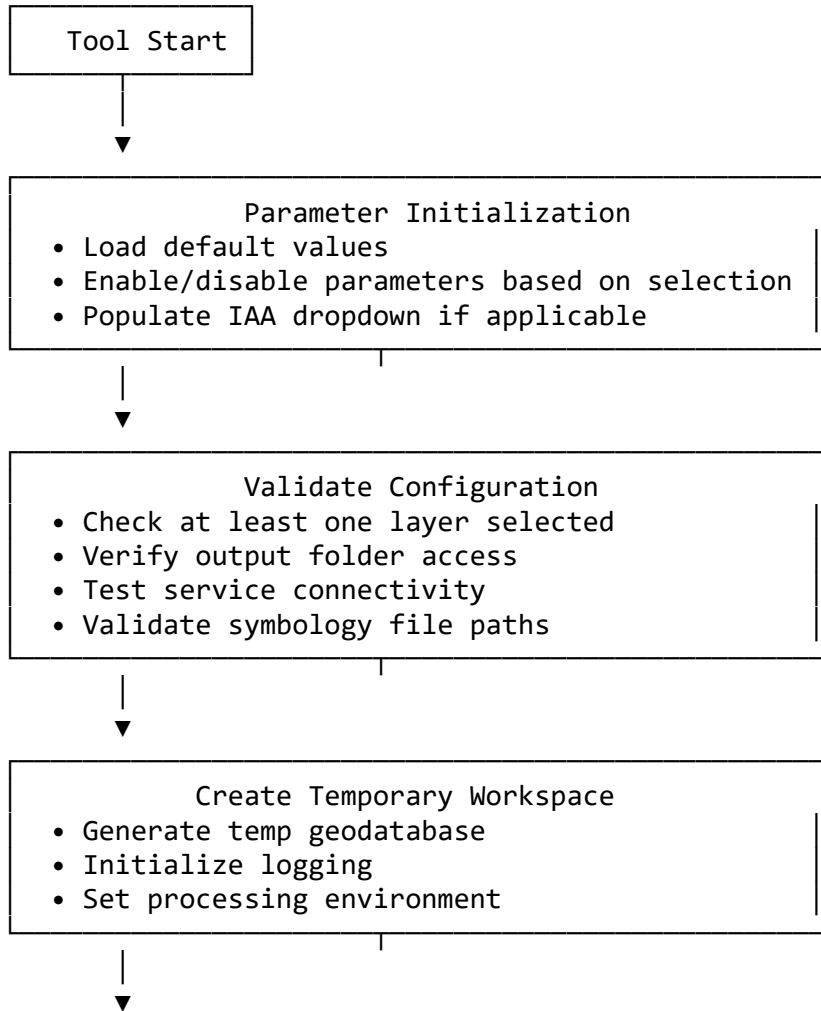
Best Practices: • Designate primary operator for daily exports

- Create user-specific symbology folders
- Implement version control for custom symbology
- Maintain centralized log repository
- Schedule staggered processing times

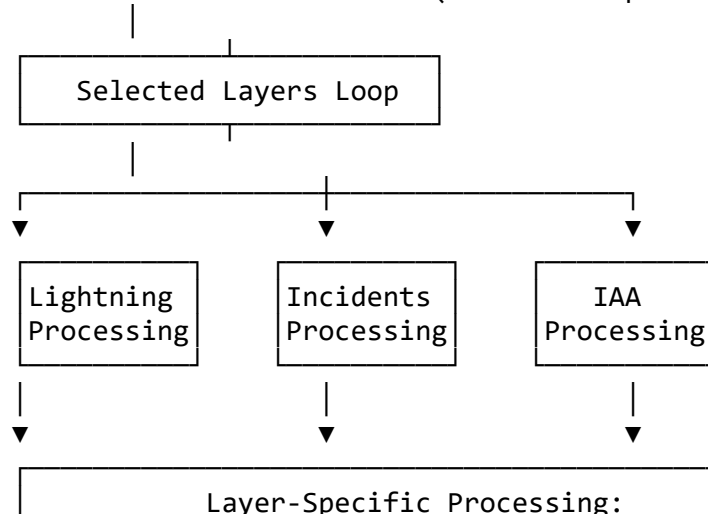
Processing Flow Diagram



PHASE 1: INITIALIZATION & VALIDATION

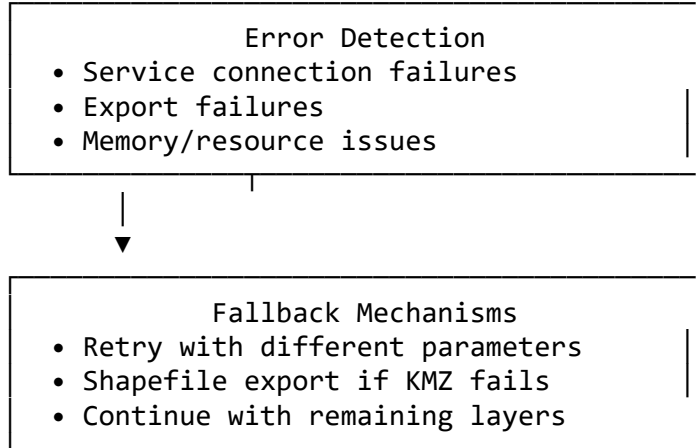


PHASE 2: LAYER PROCESSING (Parallel Capable)

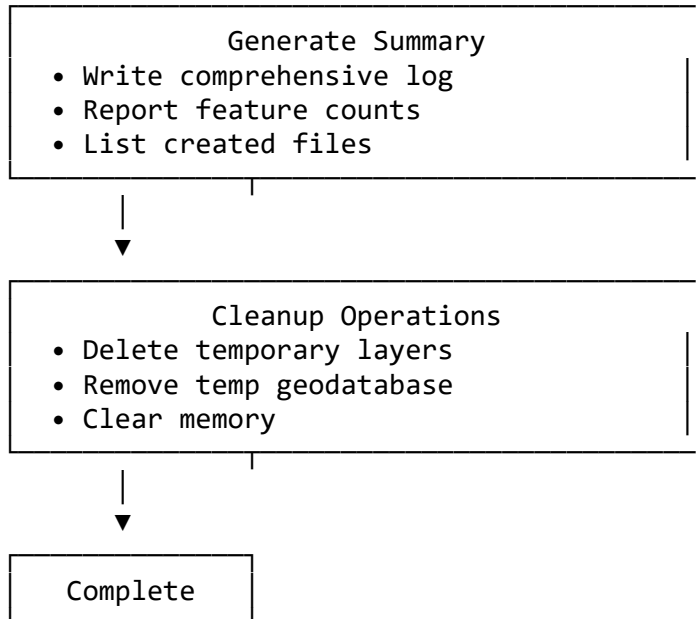


1. Service Query (with SQL filters)
2. Spatial Filtering (if applicable)
3. Feature Limiting (if applicable)
4. Symbology Application
5. KMZ Export (with fallback logic)

PHASE 3: ERROR HANDLING & RECOVERY



PHASE 4: COMPLETION & CLEANUP



KEY DECISION POINTS:

- Layer Selection → Determines processing path
- Mask Provided → Triggers spatial filtering
- State Selection → Applies SQL filtering

- IAA Filter → Modifies query parameters
- Export Failure → Initiates fallback sequence

SECTION IV - OUTPUT FILES AND DATA MANAGEMENT

File Naming Conventions

The tool implements intelligent naming conventions that clearly identify data currency, scope, and content:

Standard Format: YYYYMMDD_LayerName_State.kmz

IAA-Specific Format: YYYYMMDD_IAA_AOI_IncidentName_State.kmz

Component Definitions:

- **YYYY:** Four-digit year
- **MM:** Two-digit month (01-12)
- **DD:** Two-digit day (01-31)
- **LayerName:** Descriptive layer identifier - Lightning24h: 24-hour lightning strikes - Lightning72h: 72-hour lightning strikes - ThermalHotspots: VIIRS satellite detections - Incidents: Active fire points - Perimeters: Fire boundaries - IAA_AOI: Aviation mission areas
- **IncidentName:** Specific incident for IAA (cleaned for filesystem)
- **State:** Geographic identifier - WA: Washington only - OR: Oregon only - WAOR: Both states processed
- **Extension:** - .kmz: Primary output format - _SHP folder: Fallback shapefile output

Example Outputs:

20250829_Lightning24h_WA.kmz
20250829_Lightning72h_WAOR.kmz
20250829_ThermalHotspots_WA.kmz
20250829_Incidents_OR.kmz
20250829_Perimeters_WAOR.kmz
20250829_IAA_AOI_RidgeFire_WA.kmz
20250829_IAA_AOI_AllIncidents_WA.kmz

Log File Format: WFGIS_Export_Log_YYYYMMDD_HHMM.txt

Example: WFGIS_Export_Log_20250829_1423.txt

Shapefile Fallback Naming: YYYYMMDD_State_LayerName_SHP/

Directory contains: - Layer.shp (geometry) - Layer.shx (spatial index) - Layer.dbf (attributes) - Layer.prj (projection) - Layer.cpg (encoding)

Output Types and Formats

Primary Output - KMZ Files

KMZ (Keyhole Markup Language Zipped) files optimized for Google Earth:

Content Structure: • **Spatial Data:** Complete feature geometries in WGS 1984

- **Attributes:** All source fields preserved
- **Symbology:** DNR-standard styling embedded
- **Metadata:** Processing timestamp and parameters

Layer-Specific Optimizations:

Lightning Layers: - Point features with temporal attributes - Strike time preserved for analysis - Symbology reflects strike age - Clustering disabled for accuracy

Thermal Hotspots: - Confidence levels maintained - Detection timestamp included - Satellite pass information preserved - Temperature anomaly values retained

Incidents: - IncidentName field for labeling - Size and status attributes - Discovery date/time preserved - Cause information when available

Perimeters: - Transparency preserved (40-60%) - High-resolution edge rendering - Acre calculations included - Growth rate attributes maintained

IAA AOI: - Mission_Status field preserved - Aircraft assignment visible - Incident linkage maintained - Request timestamps included

Fallback Output - Shapefile Format

Automatic shapefile creation when KMZ export fails:

Directory Structure:

```
20250829_WA_Perimeters_SHP/  
├── temp_Perimeters.shp  
├── temp_Perimeters.shx  
├── temp_Perimeters.dbf  
├── temp_Perimeters.prj  
└── temp_Perimeters.cpg
```

Advantages: • Universal GIS compatibility

- Preserves all attributes
- Suitable for further analysis
- No symbology limitations

Limitations: • Larger file size (uncompressed)

- Requires GIS software to view

- No embedded styling
- Multiple files to manage

Log File Documentation

Comprehensive processing record with timestamped entries:

Log Sections:

```
===== WFGIS Export Started =====
10:23:45 - Selected layers for export: 24h Lightning, Incidents, Perimeters
10:23:46 - [Lightning24h] Processing...
10:23:48 - [Lightning24h] Service connection established
10:23:52 - [Lightning24h] Spatial filter applied - 12,847 features reduced to
8,432
10:23:53 - [Lightning24h] Limiting to 15000 features...
10:24:03 - [Lightning24h] Applying symbology: 24h_Lightning.lyrx
10:24:15 - [Lightning24h] Successfully exported KMZ: 20250829_Lightning24h_WA
.kmz (2.3MB)
10:24:16 - [Incidents] Processing...
10:24:18 - [Incidents] Applying state filter: WA
10:24:19 - [Incidents] State filtered count: 47
10:24:20 - [Incidents] Label field prepared
10:24:25 - [Incidents] Successfully exported KMZ: 20250829_Incidents_WA.kmz (
156KB)
10:24:26 - [Perimeters] Processing...
10:24:28 - [Perimeters] State filtered count: 12
10:24:35 - [Perimeters] Successfully exported KMZ: 20250829_Perimeters_WA.kmz
(1.8MB)
10:24:36 - Cleaned up temporary data
10:24:36 - Processed 3 selected layer(s): Lightning24h, Incidents, Perimeters
===== WFGIS Export Completed =====
```

Critical Information Captured: • Processing start/end times

- Feature counts at each stage
- Filtering effects
- File sizes
- Error messages and warnings
- Recovery actions taken
- Total processing duration

Quality Control

Automated Validation Procedures

The tool implements multiple validation layers:

Pre-Processing Validation: • Service connectivity verification

- Parameter completeness check
- Symbolology file accessibility
- Output directory permissions
- Mask geometry validity

During Processing: • Feature count monitoring

- Memory usage tracking
- Network timeout detection
- Spatial reference validation
- Attribute schema verification

Post-Processing Validation: • Output file size verification (>1KB minimum)

- KMZ structure validation
- Feature count reconciliation
- Log file completeness
- Cleanup verification

Manual Verification Procedures

Post-Processing Checklist:

1. File Verification

- All selected layers have outputs
- File sizes within expected ranges:
 - Lightning: 1-10MB typical
 - Incidents: 100KB-1MB typical
 - Perimeters: 1-5MB typical
 - Thermal: 500KB-3MB typical
 - IAA: 50KB-500KB typical
- Log file present and complete

2. Google Earth Testing

- KMZ files open without errors
- Symbolology displays correctly
- Labels visible where expected
- Transparency preserved (perimeters)
- Popups show attributes
- Spatial accuracy verified

3. Data Currency

- Lightning within time windows
- Incidents show current status
- Perimeters are recent

- Thermal detections current
- IAA assignments up-to-date

4. Spatial Accuracy

- Features in correct locations
- Mask filtering applied correctly
- State boundaries respected
- No obvious spatial errors

Common Quality Issues and Solutions

Issue: Missing Labels in Google Earth - Cause: Complex labeling expressions not supported - Solution: Tool creates simplified LABEL_TEXT field - Verification: Check IncidentName field exists

Issue: Blurry Perimeter Edges - Cause: Composite rendering in KML export - Solution: Tool uses NO_COMPOSITE parameter - Fallback: Multiple export attempts with different settings

Issue: Incomplete Lightning Data - Cause: Feature limit reached - Solution: Check log for limiting messages - Action: Increase Max Lightning Features if needed

Issue: IAA Dropdown Shows “Error loading incidents” - Cause: Service connection failure - Solution: Verify network connectivity - Workaround: Use “All Incidents” option

Issue: Large File Sizes - Cause: No feature limiting applied - Solution: Apply mask or reduce feature limits - Balance: Coverage vs. file size

SECTION V - TROUBLESHOOTING AND OPTIMIZATION

Common Issues

Issue: Tool Fails to Initialize

Symptoms: - Parameter dialog doesn't open - Error message about missing toolbox - ArcGIS Pro crashes on tool launch

Solutions: 1. Verify Python environment: `python import sys print(sys.version)`
Should be 3.x 2. Re-add toolbox to project 3. Check file permissions on .pyt file 4. Run ArcGIS Pro as administrator 5. Clear ArcGIS Pro cache and restart

Issue: “At least one layer must be selected” Error

Symptoms: - Error appears even with selections made - Cannot proceed with tool execution

Solutions: 1. Ensure at least one checkbox is checked 2. Close and reopen tool dialog 3. Check for parameter corruption: - Clear all selections - Re-select desired layers 4. Verify tool version matches documentation

Issue: Service Connection Failures

Error Messages: - “Service connection failed” - “Cannot connect to [service URL]” - Timeout errors in log

Solutions: 1. Test service URLs in browser: - Copy URL from Appendix A - Paste in browser - Should see REST endpoint 2. Check firewall settings: - Allow services3.arcgis.com - Allow services9.arcgis.com 3. Verify network proxy settings 4. Try again during off-peak hours 5. Contact IT for firewall exceptions

Issue: KMZ Export Fails, Creates Shapefile Instead

Log Messages: - “KMZ export failed” - “Falling back to shapefile export”

Common Causes & Solutions:

1. **Invalid Characters in Attributes**
 - Remove special characters from data
 - Tool attempts automatic cleaning
2. **Geometry Errors**
 - Run Repair Geometry on source data
 - Simplify complex polygons
3. **File Path Too Long**
 - Use shorter output directory path
 - Reduce incident name length for IAA
4. **Insufficient Permissions**
 - Verify write access to output folder
 - Check available disk space

Issue: IAA Dropdown Not Populating

Symptoms: - Dropdown shows “Error loading incidents” - Only “All Incidents” option available

Solutions: 1. Verify IAA service accessibility 2. Check network connectivity 3. Ensure Export IAA is checked 4. Wait 10-15 seconds for population 5. Use “All Incidents” as workaround

Issue: Symbolology Not Applied

Log Messages: - “Could not apply symbolology” - “Symbolology file not found”

Solutions: 1. Verify .lyrx/.lyr file exists 2. Check file path accessibility 3. Ensure symbology matches data schema 4. Clear parameter to use defaults 5. Create new symbology file from current data

Issue: Lightning Data Missing or Incomplete

Symptoms: - Very few or no lightning features - Unexpected gaps in coverage

Potential Causes & Solutions:

1. **No Recent Lightning Activity**
 - Check weather patterns
 - Verify with external sources
2. **Mask Too Restrictive**
 - Review mask boundary
 - Try without mask to verify
3. **Feature Limit Too Low**
 - Increase Max Lightning Features
 - Check log for limiting messages
4. **Service Data Lag**
 - WFIGS updates every 15-30 minutes
 - Try again after waiting

Issue: Processing Takes Excessive Time

Symptoms: - Tool runs for >30 minutes - ArcGIS Pro becomes unresponsive

Solutions: 1. Reduce selected layers 2. Apply spatial mask 3. Lower feature limits 4. Process states separately 5. Close other applications 6. Use local output directory 7. Check network bandwidth

Performance Optimization

System-Level Optimizations

Hardware Upgrades: - RAM: 16GB minimum, 32GB optimal - Storage: SSD for temp files and output - CPU: Multi-core processor (4+ cores) - Network: Wired connection preferred

Software Configuration:

```
# In Python window, set environment
arcpy.env.parallelProcessingFactor = "100%"
arcpy.env.processorType = "GPU" # If available
```

ArcGIS Pro Settings: 1. Options > Display > Hardware Acceleration: Enable 2. Options > Processing > Background: Enable 3. Clear cache regularly: Options > Display > Clear Cache

Query Optimization Strategies

Selective Layer Processing: - Only select needed layers - Process critical layers first - Schedule comprehensive exports off-hours

Efficient Filtering:

```
-- Optimized incident query
WHERE IncidentSize >= 1
      AND FireDiscoveryDateTime > CURRENT_TIMESTAMP - 30
      AND ControlDateTime IS NULL
-- Indexes on these fields improve performance
```

Spatial Filtering Best Practices: - Simplify mask polygons (<1000 vertices) - Use single-part features - Project mask to WGS 1984 beforehand - Avoid complex boundaries when possible

Network Optimization

Service Request Management: - Process during off-peak hours (early morning) - Implement retry logic for failures - Cache service responses when possible

Bandwidth Considerations: - Lightning data: ~5-10MB download - Incidents: ~1-2MB download - Perimeters: ~2-5MB download - Plan for 20-30MB total transfer

Processing Time Benchmarks

Typical processing times by configuration:

Configuration	Layers	Features	Time
Minimal	Lightning 24h only	5,000	2-3 min
Quick Intel	Lightning + Incidents	10,000	4-6 min
Standard	All except IAA	15,000	8-12 min
Comprehensive	All layers	25,000	15-20 min
Maximum	All, no limits	Unlimited	30-60 min

Memory Management

Feature Limiting Strategy:

```
# Tool uses cursor iteration to manage memory
with arcpy.da.SearchCursor(fc, fields) as cursor:
    for row in cursor:
```

```
if count >= max_features:
    break
# Process feature
count += 1
```

Temporary File Management: - Tool creates temp geodatabase - Automatic cleanup on completion - Manual cleanup if tool fails:

```
import tempfile
import os
temp_dir = tempfile.gettempdir()
# Look for wfgis_temp_*.gdb and delete
```

Best Practices

Operational Planning

Daily Export Schedule: - Morning (0600): Comprehensive overnight intelligence - Midday (1200): Incident and perimeter updates - Evening (1800): Final operational picture - As-needed: Rapid exports for new starts

Layer Selection Guidelines: - Initial Attack: Lightning 24h + Incidents + Thermal - Daily Briefing: All layers except 72h lightning - Investigation: Both lightning layers + Incidents - IAA Mission: IAA + relevant operational layers

Data Management

File Organization:

Recommended Directory Structure:

```
/WFGIS_Exports/
├── /Daily_Routine/
│   ├── /2025-08/
│   │   ├── /20250829/
│   │   │   ├── Morning/
│   │   │   ├── Afternoon/
│   │   │   └── Evening/
│   └── /Incident_Specific/
│       ├── /RidgeFire/
│       ├── /ValleyComplex/
│       └── /BlueMtnRx/
├── /IAA_Missions/
│   ├── /N565E/
│   └── /N216KQ/
├── /Archives/
│   └── /2025_Q3/
└── /Logs/
    └── /2025-08/
```


Retention Policy: - Daily exports: 30 days - Incident-specific: Duration + 90 days - IAA missions: Duration + 60 days - Logs: 1 year minimum - Archives: Annual compression

Quality Assurance

Pre-Processing Verification: 1. Review previous day's log for issues 2. Check service status (Appendix A URLs) 3. Verify mask features if used 4. Confirm symbology files accessible 5. Test with single layer first if issues suspected

Post-Processing Validation: 1. Open log file immediately 2. Verify expected feature counts 3. Test one KMZ in Google Earth 4. Check file sizes for anomalies 5. Confirm spatial extent correct

Documentation Requirements: - Log significant parameter changes - Document custom symbology use - Note any service outages - Record processing anomalies - Track average processing times

Multi-User Coordination

Processing Assignments: - Designate primary operator per shift - Establish backup operators - Create standardized parameter sets - Document custom configurations

Communication Protocols:

Email Template for Export Notification:

Subject: WFGIS Export Complete - [DATE] [TIME]

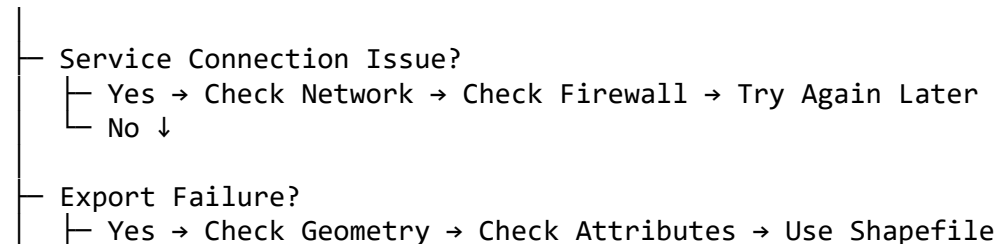
Layers Processed: [List]
State Coverage: [WA/OR/Both]
Special Filters: [Mask/IAA filter]
Output Location: [Path]
Processing Time: [Duration]
Issues/Notes: [Any anomalies]

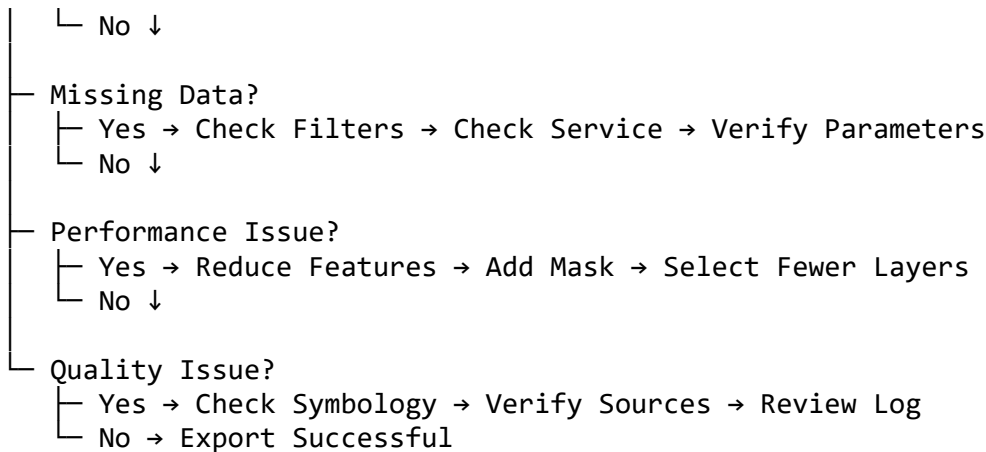
Files Available:

- [List of KMZ files with sizes]
- Log: [Filename]

Troubleshooting Decision Tree

Problem Detected





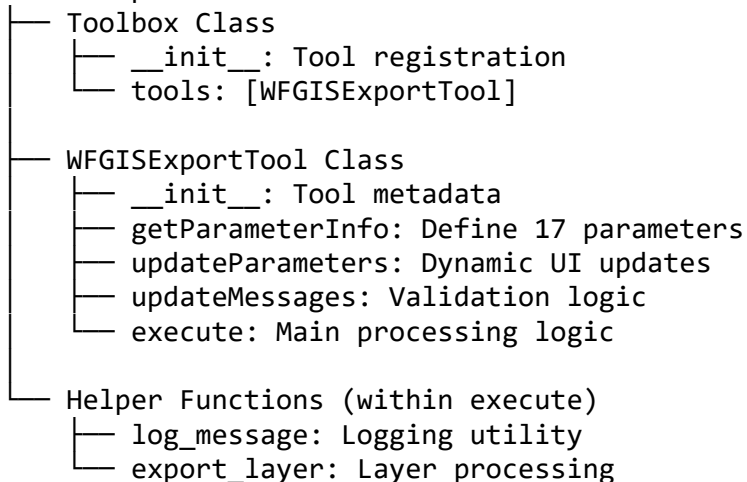
SECTION VI - TECHNICAL REFERENCE

Code Architecture

Tool Structure

The WFGIS Export Tool is implemented as a Python Toolbox (.pyt) with object-oriented design:

Main Components:



Parameter Management

Parameters are indexed 0-16 with specific roles:

Index	Parameter	Type	Role
0-5	Layer selections	Boolean	Enable/disable processing
6	State	String	Geographic filtering

Index	Parameter	Type	Role
7	IAA Filter	String	Incident filtering
8	Mask	Feature Layer	Spatial filtering
9	Output Folder	Folder	File destination
10	Max Lightning	Integer	Performance control
11-16	Symbology	File	Visual styling

Processing Pipeline

Sequential processing with error isolation:

1. Initialization Phase

- Parameter validation
- Service connectivity check
- Workspace creation
- Logging initialization

2. Layer Processing Loop

- Iterate selected layers
- Apply appropriate filters
- Export with fallback logic
- Isolate errors per layer

3. Cleanup Phase

- Remove temporary data
- Finalize logging
- Report summary

Service Endpoints and Data Sources

Lightning Services

24-Hour Lightning: - URL:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Lightning_24h/FeatureServer/0 - Update Frequency: 15-30 minutes - Source: Lightning Detection Network - Coverage: CONUS + Alaska - Attributes: Strike time, amplitude, polarity

72-Hour Lightning: - URL:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Lightning_72h/FeatureServer/0 - Update Frequency: 30 minutes - Archive Depth: Rolling 72 hours - Use Case: Pattern analysis, cause determination

Fire Intelligence Services

Active Incidents: - URL:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Incident_Locations_YearToDate_Internal/FeatureServer/0 - Update Frequency: Real-time - Source: IRWIN/InciWeb - Filters Applied: sql GACC = 'NWCC' POOState IN ('US-WA','US-OR') IncidentTypeCategory IN ('WF','RX') IncidentSize >= 1

Fire Perimeters: - URL:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Interagency_Perimeters_Current/FeatureServer/0 - Update Frequency: Daily - Source: NIFC/GeoMAC - Minimum Size: 50 acres

Satellite Services

VIIRS Thermal Hotspots: - URL:

https://services9.arcgis.com/RHVPKKiFTONKtxq3/arcgis/rest/services/Satellite_VIIRS_Thermal_Hotspots_and_Fire_Activity/FeatureServer/0 - Update Frequency: 3-4 hours - Satellites: NOAA-20, Suomi-NPP - Resolution: 375 meters - Attributes: Confidence, temperature, detection time

Aviation Services

IAA Mission Requests: - URL:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/2025_Mission_Request_IAA_Only/FeatureServer/0 - Update Frequency: Real-time - Aircraft: N565E, N216KQ - Filter: Mission_Status = 'Assigned'

Configuration Details

Default File Paths

Base Directory:

J:\Aviation_ops\Air Program\7-Chief Pilot\Kodiak 100\Fixed Wing (FW)\MSO\MSO GIS\2025\

Symbology Files:

```
└─ 2025_kml_file_conversions\  
    └─ Tools\  
        └─ Layer Files\  
            └─ 24h_Lightning.lyrx  
            └─ 72h_Lightning.lyrx  
            └─ Thermal_Hotspots.lyrx  
            └─ WFIGS_Incidents.lyrx  
            └─ WFIGS_Perimeters.lyrx  
            └─ IAA_2025_Mission_Request.lyrx
```

Output Directory:

```
└─ 2025_kml_file_conversions\  
    └─ KML_Files\  
        └─ WFIGS_Daily_Exports\  
            └─
```

Environment Settings

Optimal arcpy environment configuration:

```
arcpy.env.overwriteOutput = True  
arcpy.env.addOutputsToMap = False  
arcpy.env.workspace = temp_gdb  
arcpy.env.scratchWorkspace = temp_gdb
```

Error Handling Strategy

Three-tier approach: 1. **Try Primary Method** - Standard KML export with full parameters 2.

Fallback to Minimal Parameters - Simplified export without advanced options 3.

Emergency Shapefile Export - Guaranteed output method - Preserves all data

Maintenance and Updates

Version Control

Track changes in header comments:

```
"""  
WFIGS Export Tool  
Version: 2.0  
Date: August 2025  
Author: DNR Aviation GIS Team  
Changes:  
- Added layer selection checkboxes  
- Integrated thermal hotspot support  
- Added IAA mission filtering  
- Enhanced error handling  
"""
```

Service Monitoring

Regular verification schedule: 1. Weekly: Test all service endpoints 2. Monthly: Verify SQL filters still valid 3. Quarterly: Review symbology files 4. Annually: Update service URLs if changed

Update Procedures

1. Minor Updates (bug fixes):

- Update code in .pyt file
- Test with sample data
- Deploy to users

- No manual update needed

2. Major Updates (new features):

- Update code and documentation
- Create migration guide
- Train users on changes
- Archive previous version

Debugging Techniques

Enable verbose logging:

Add to execute method

`DEBUG_MODE = True` *# Set to False for production*

if `DEBUG_MODE`:

```
    log_message(f"Debug: Parameter values: {[p.value for p in parameters]}")
    log_message(f"Debug: Service URL: {service_url}")
    log_message(f"Debug: SQL: {sql_clause}")
```

Common Customizations

Adding New Layers: 1. Add checkbox parameter 2. Add symbology parameter 3. Add service configuration 4. Update export_layer logic 5. Test thoroughly

Modifying Filters:

Example: Change incident size threshold

```
services["Incidents"]["sql"] = \
    f"... AND IncidentSize >= 10 ..." # Changed from 1
```

Custom Output Formats:

Add GeoJSON export option

```
arcpy.conversion.FeaturesToJSON(
    temp_fc,
    out_json,
    format_json="FORMATTED"
)
```

GLOSSARY

AOI - Area of Interest; geographic boundary for focused analysis

DNR - Washington State Department of Natural Resources

Fallback - Alternative processing method when primary fails

Feature Layer - In-memory representation of geographic data

GACC - Geographic Area Coordination Center

GeoMAC - Geospatial Multi-Agency Coordination (legacy, replaced by NIFC)

IAA - Interagency Aviation Agreement

IMT - Incident Management Team

InciWeb - Incident Information System (public interface)

IRWIN - Integrated Reporting of Wildland-Fire Information

KML/KMZ - Keyhole Markup Language (KMZ is compressed)

MMA - Multi Mission Aircraft program

NIFC - National Interagency Fire Center

NWCC - Northwest Coordination Center

POO - Point of Origin

REST - Representational State Transfer (web service type)

RX - Prescribed fire/burn

Symbology - Visual representation rules for map features

VIIRS - Visible Infrared Imaging Radiometer Suite

WF - Wildfire

WFIGS - Wildland Fire Interagency Geospatial Services

WGS 1984 - World Geodetic System 1984 (coordinate system)

WUI - Wildland-Urban Interface

APPENDIX A - SERVICE URL REFERENCE

Test these URLs in a browser to verify connectivity:

Lightning Services: - 24-Hour:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Lightning_24h/FeatureServer/0 - 72-Hour:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Lightning_72h/FeatureServer/0

Fire Intelligence: - Incidents:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Incident_Locations_YearToDate_Internal/FeatureServer/0 - Perimeters:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/WFIGS_Interagency_Perimeters_Current/FeatureServer/0

Satellite Data: - Thermal:

https://services9.arcgis.com/RHVPKKiFTONKtxq3/arcgis/rest/services/Satellite_VIIRS_Thermal_Hotspots_and_Fire_Activity/FeatureServer/0

Aviation: - IAA:

https://services3.arcgis.com/T4QMspbfLg3qTGWY/arcgis/rest/services/2025_Mission_Request_IAA_Only/FeatureServer/0

APPENDIX B - CHANGE LOG

Version 2.0 (August 2025)

Major Features: - Added individual layer selection checkboxes (6 total) - Integrated VIIRS thermal hotspot support - Added IAA Area of Interest with dynamic filtering - Enhanced user interface with parameter grouping

Improvements: - Intelligent parameter enabling/disabling - Enhanced error handling with multiple fallback methods - Improved KML export parameters for each layer type - Better logging with selected layer tracking - Optimized cleanup procedures

Bug Fixes: - Fixed perimeter transparency issues - Resolved incident labeling in Google Earth - Corrected memory leaks in layer cleanup - Fixed IAA incident name handling

Version 1.0 (May 2025)

Initial Release: - Basic lightning, incident, and perimeter export - State filtering for WA/OR - Mask support for lightning - Standard DNR symbology - KMZ export with shapefile fallback

For Technical Support: Contact: DNR Aviation GIS Team

Email: aviation.gis@dnr.wa.gov

Phone: (360) 902-1000

Documentation Maintenance: Last Updated: August 2025

Next Review: November 2025

Document Version: 2.0

END OF DOCUMENT

This manual represents the complete documentation for the WFGIS Export Tool Version 2.0. For updates and additional resources, please contact the DNR Aviation GIS Team.
